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Financial Analytics Lab

GROUP 11

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01 # Answers

[Link to PYTHON CODE](#)

Q1.

Countries chosen are as follows (all are part of the G7 group)

- *Canada*
- *France*
- *Germany*
- *Italy*
- *Japan*

A unit root is a feature of a time series where the process is non-stationary, meaning the mean and variance of the series are not constant over time. In particular, a unit root implies that the series has a stochastic trend and does not revert to a constant mean over time.

Mathematically, a time series Y_t is said to have a unit root if it satisfies the following equation: $Y\_t = Y\_(t-1) + \epsilon\_t$

where ϵ_t is a white noise error term. This equation indicates that the current value of the series depends on the previous value plus some random shock.

In financial time series analysis, the presence of a unit root is of particular interest because it implies that the series is not mean-reverting and may exhibit persistent trends. Understanding whether a financial time series has a unit root is crucial for various applications, including:

1. **Forecasting**: Knowing whether a series has a unit root helps in choosing appropriate forecasting models. For instance, if a series has a unit root, traditional forecasting methods that assume stationarity may not be appropriate, and alternative methods like autoregressive integrated moving average (ARIMA) models or vector autoregression (VAR) models may be more suitable.
2. **Risk Management**: In finance, understanding the behavior of asset prices and returns is essential for risk management. A series with a unit root may indicate that shocks to the system can have long-lasting effects, affecting risk assessments and portfolio management strategies.
3. **Cointegration Analysis**: Unit root testing is often used in cointegration analysis, which examines the long-term relationship between two or more time series. Cointegration implies that although individual series may have unit roots, linear combinations of these series are stationary. This concept is widely applied in pairs trading, asset pricing models, and risk arbitrage strategies.
4. **Market Efficiency**: The presence of unit roots in financial time series can also provide insights into market efficiency. For example, if stock prices follow a random walk (i.e., have a unit root), it suggests that past prices do not contain useful information for predicting future prices, supporting the efficient market hypothesis.

Answers

Q1.

Code results for unit root test for each of the 5 countries we have chosen:

Country	ADF Statistic	p-value
Canada	-0.621	0.866
France	-0.591	0.873
Germany	-3.682	0.004
Italy	-4.628	0.000
Japan	2.003	0.999

Based on the above tabulated results from our Colab code file:

- Due to very less p-value of Germany and Italy (<0.05)
- Therefore these values are significant
- Null Hypothesis: Unit root exists
- Therefore for Germany and Italy we can reject the unit root and therefore they are STATIONARY.

Q2.

Lag length in the context of unit root tests refers to the number of lagged differences included in the test equation. In time series analysis, when testing for a unit root using methods like the Augmented Dickey-Fuller (ADF) test, it's common to include lagged differences of the variable being tested. This is done to capture any serial correlation present in the data.

In the provided prompts, each country's data is subjected to the ADF test, which tests for the presence of a unit root in the time series data. The ADF test involves specifying a model with lagged differences of the variable being tested. The lag length determines how many lagged differences are included in the model.

Regarding the impact of lag length on unit root tests:

1. **Choosing the appropriate lag length**: Selecting the appropriate lag length is crucial in unit root tests. If the lag length is too short, important serial correlation might be omitted, leading to biased results. Conversely, if the lag length is too long, it can introduce unnecessary complexity and reduce the power of the test.
2. **Automatic lag selection methods**: In practice, researchers often employ automatic lag selection methods, such as Akaike Information Criterion (AIC) or Bayesian Information Criterion (BIC), to determine the optimal lag length. These methods aim to strike a balance between model complexity and goodness of fit.
3. **Interpretation of results**: When interpreting the results of unit root tests, it's important to consider the lag length chosen for the analysis. Different lag lengths might lead to different conclusions regarding the presence or absence of a unit root in the data.

Answers

To analyze the results of the lag length on the unit root test we proceed with the following steps:

- We take 4 lag values as lag_length = [3,4,5,6]
- We then iterate the whole process to find the results as follows:

Lag Length	Country	ADF Statistic	p-value
3	Canada	-2.603	0.092
3	France	-0.591	0.873
3	Germany	-3.682	0.004
3	Italy	-4.628	0.000
3	Japan	1.134	0.995
4	Canada	-1.089	0.719
4	France	-0.591	0.873
4	Germany	-1.244	0.654
4	Italy	-4.628	0.000
4	Japan	1.134	0.995
5	Canada	-0.838	0.808
5	France	-0.591	0.873
5	Germany	-3.682	0.004
5	Italy	-4.628	0.000
5	Japan	1.667	0.998
6	Canada	-0.621	0.866
6	France	-0.591	0.873
6	Germany	-3.682	0.004
6	Italy	-4.628	0.000
6	Japan	1.667	0.998

Observations:

- **Only in case of lag length = 4 , ITALY is the only country with the stationary FDI values**
- **Rest in all other cases lag length = 3,5,6 , ITALY AND GERMANY are the two countries with the stationary FDI values.**

Answers

Q3. A structural break denotes a sudden and enduring alteration in the relationship among variables within a time series or in the properties of the time series itself. Such breaks can significantly influence the analysis and forecasting of time series data by signaling a fundamental change in the process generating the data.

In the realm of unit root tests and the examination of non-stationarity, structural breaks carry particular importance. Failing to accommodate for a structural break may result in a unit root test erroneously concluding that a series is non-stationary due to the shift introduced by the break, rather than stemming from a genuinely unit root process.

To address the impact of structural breaks on unit root testing, several approaches can be adopted. Modified unit root tests such as the Zivot-Andrews test and techniques involving pre-testing for breaks are utilized. These methodologies integrate the possibility of structural breaks, yielding more precise determinations regarding the presence of unit roots.

Considering the dataset from Problem 1, if economic or policy shifts have induced structural breaks in the data, it is advisable to account for these breaks during unit root tests. Neglecting potential breaks may lead to erroneous conclusions regarding the stationarity of the series.

Answers

Q4. Cointegration is a statistical concept used in econometrics to describe the long-term equilibrium relationship between two or more time series variables. It is particularly relevant in the analysis of non-stationary time series data.

In simpler terms, cointegration implies that while individual time series may not be stationary (i.e., their mean and variance change over time), there exists a combination of these variables that is stationary. This means that even though the individual series may exhibit trends or drifts, the combination of these series moves around a constant mean or exhibits a stable relationship over time.

Cointegration is commonly associated with the Error Correction Model (ECM), which models the short-term dynamics of variables adjusting towards their long-term equilibrium relationship. It's widely used in macroeconomics, finance, and other fields where researchers are interested in understanding the relationship between variables that may not move together in the short term but are linked in the long run.

Country Pair	Cointegration Statistic	p-value	Cointegration Status
Canada - France	-2.4309	0.3104	Not cointegrated
Canada - Germany	-4.1624	0.0042	Cointegrated
Canada - Italy	-1.9593	0.5497	Not cointegrated
Canada - Japan	-3.8174	0.0128	Cointegrated
France - Germany	-1.9871	0.5353	Not cointegrated
France - Italy	-0.4035	0.9705	Not cointegrated
France - Japan	-1.3633	0.8098	Not cointegrated
Germany - Italy	-3.0085	0.1082	Not cointegrated
Germany - Japan	-5.1778	0.0001	Cointegrated
Italy - Japan	-5.2124	0.0001	Cointegrated

Tabular compilation of the code results